

Statistical Physics

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24 de febrero de 2026

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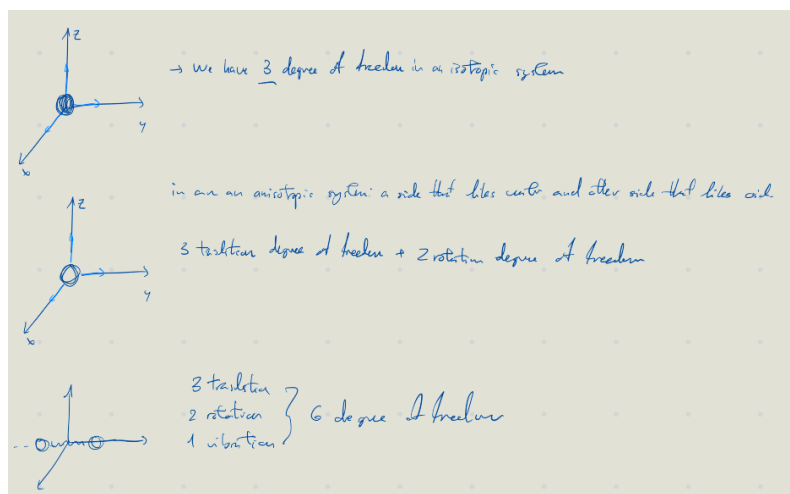
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Capítulo 1

Microscopic description

Statistical physics serves as the bridge that connects the microscopic world of atoms and molecules to the macroscopic phenomena observed in our everyday lives. At its core, this field is concerned with unraveling the mysteries of nature by understanding how countless microscopic particles, when aggregated, give rise to the familiar and predictable behaviour of macroscopic systems. This chapter provides us with the tools and knowledge necessary to comprehend the underlying dynamics of particles at the smallest scales. Here, we shift our focus from macroscopic observables like pressure and temperature to study individual particles – their motions, interactions, and the probabilistic nature of their behaviour. As we walk through the microscopic world, we will uncover the principles governing the behaviour of molecules and atoms. We will decipher the significance of concepts such as phase space, Hamiltonian mechanics, and quantum states, all of which play crucial roles in painting a vivid portrait of the microscopic universe. By the end of this chapter, you will possess a profound understanding of how microscopic details pave the way for macroscopic phenomena.

1.1. Classical Description



1. An isotropic atom has no rotational degrees of freedom, but only three translational degrees of freedom. Since any rotation of a perfectly symmetrical sphere constitutes an identity transformation, the orientation does not alter the particle's interaction potential. Consequently, the system's energy remains invariant under rotation, meaning no thermal energy can be coupled to this motion, and the rotational degrees of freedom are effectively non-existent in the statistical count.
2. An anisotropic atom of two sides, one like water and one like oil has 2 rotational degrees of freedom and the three translational degrees of freedom. In this scenario, where the northern and southern hemispheres exhibit distinct interaction potentials, the perfect spherical symmetry is broken and replaced by axial symmetry. A rotation around the Z-axis (the axis of symmetry) leaves the configuration invariant, meaning it does not change the physical state. However, rotations around the X and Y axes alter the orientation of the hemispheres relative to their surroundings, creating distinguishable states. Consequently, we account for exactly two rotational degrees of freedom.
3. A diatomic molecule has 3 translational, 2 rotational and 1 vibrational (due to the distance between both particles) degrees of freedom.

Just for curiosity

Degrees of Freedom (s)	A (Atom $\implies$ N=1)	Linear Molecule	Non-linear Molecule
Translational	3(N)	3	3
Rotation	0	2	3
Vibration	0	3N-5	3N-6
Total	3(N)	3N	3N

The degree of freedom is always $3N$.

Nota

In this chapter we will only treat particles as **simple spheres**.

- Let's consider a mechanical system composed of a certain number of particles that has a total of **s degrees of freedom**. To specify the **dynamic state** of this system, it is necessary to know the **positions and velocities** of all the particles that comprise it. Specifically, a total of s position coordinates and s corresponding momenta associated with these coordinates.

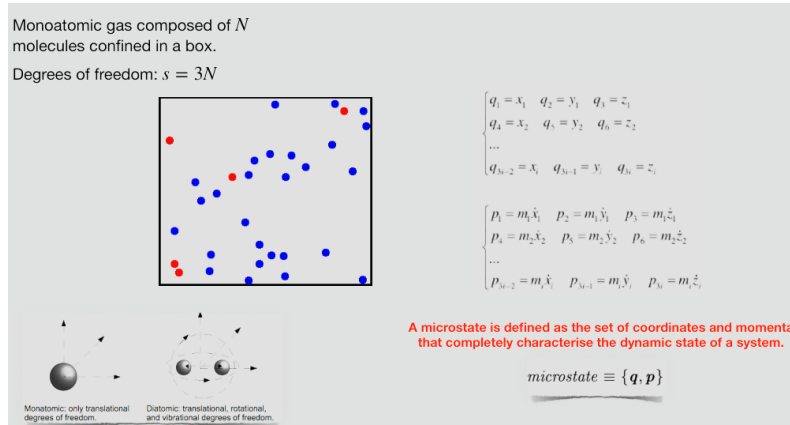
- Making use of *Analytical Mechanics*, we can employ generalised coordinates and generalised momenta:

$$\{q_1, q_2, \dots, q_s, p_1, p_2, \dots, p_s\} \equiv \{\mathbf{q}, \mathbf{p}\}$$

Generalised coordinates and momenta can correspond to the **positions** and **momenta** of the particles in the system, although they can also represent more abstract quantities such as **angles**, **wave amplitudes**, or the different **internal degrees of freedom** of a particle.

In classical physics, the description of a system often involves a fundamental and powerful concept known as phase space, here indicated as Γ . Phase space provides a comprehensive mathematical framework for understanding and predicting

the behaviour of physical systems. It is particularly essential in classical mechanics for analysing systems with complex configurations and interactions. Phase space provides physicists and engineers with a powerful tool for modelling and analysing complex systems. It allows us to visualise and study the system's evolution over time by tracing its path through the multidimensional space. This approach simplifies the prediction of a system's future behaviour, making it a fundamental concept in classical mechanics and other branches of physics. Understanding phase space is essential for tackling a wide range of problems, from the motion of celestial bodies to the behaviour of particles in a gas. It forms the basis for classical mechanics and serves as a foundation for more advanced topics in physics. At its core, phase space is a multidimensional mathematical space that offers a complete representation of the state of a physical system. Each point within phase space corresponds to a **unique state of the system**. The number of dimensions in phase space equals the number of **generalised coordinates** and **generalised momenta** required to define the system's configuration. For example, in a system with s degrees of freedom, the corresponding phase space would be a $2s$ -dimensional space. This concept allows us to track the system's evolution through all possible states by examining its trajectory in phase space.



Consider a monoatomic gas composed of N molecules confined within a box. In this case, the degrees of freedom are given by $s = 3N$. Its generalized coordinates for positions and momenta can be written as follows:

Generalised coordinates

$$\begin{cases} q_1 = x_1 & q_2 = y_1 & q_3 = z_1 \\ q_4 = x_2 & q_5 = y_2 & q_6 = z_2 \\ \dots \\ q_{3i-2} = x_i & q_{3i-1} = y_i & q_{3i} = z_i \end{cases}$$

Generalised momenta

$$\begin{cases} p_1 = m_1 \dot{x}_1 & p_2 = m_1 \dot{y}_1 & p_3 = m_1 \dot{z}_1 \\ p_4 = m_2 \dot{x}_2 & p_5 = m_2 \dot{y}_2 & p_6 = m_2 \dot{z}_2 \\ \dots & & \\ p_{3i-2} = m_i \dot{x}_i & p_{3i-1} = m_i \dot{y}_i & p_{3i} = m_i \dot{z}_i \end{cases}$$

In this representation, q_i denotes the position coordinates, while p_i corresponds to the momenta of the i -th particle within the system. These variables, combined with time, allow us to describe the complete state of the system dynamically.

Microstate Definition

A microstate is defined as the set of coordinates and momenta that completely characterise the dynamic state of a system.

$$microstate \equiv \{\mathbf{q}, \mathbf{p}\} \equiv \{q_1, q_2, \dots, p_1, p_2, \dots, p_s\} \quad (1.1)$$

1.1.1. Phase Space

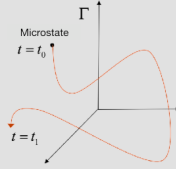
- The phase space, denoted as Γ , is defined as a $2s$ -dimensional metric space, where s is the number of degrees of freedom in the system.
- The axes of this phase space correspond to the variables $\{q_1, \dots, q_s, p_1, \dots, p_s\}$. In the phase space, the **microstate of the system is represented by a point**.
- The temporal evolution of the system results in a **trajectory** within the phase space, traversing different microstates.

Example: diatomic molecule (e.g., O_2)

$s = 3N = 3 \times 2$ degrees of freedom $\Rightarrow \Gamma$ is a space of $2s = 12$ dimensions

Dimensions:

- 3 positions of the center of mass q_x, q_y, q_z
- 3 linear momenta p_x, p_y, p_z
- 2 angular coordinates: θ and ϕ
- 2 angular momenta: p_θ and p_ϕ
- 1 stretching coordinate (distance between atoms), q_r
- 1 stretching momentum: p_r



The space of phases (or phase space), denoted as Γ , is defined as a $2s$ -dimensional metric space, where s is the number of degrees of freedom in the system. The axes of this phase space correspond to the variables $\{q_1, \dots, q_s, p_1, \dots, p_s\}$. In the phase space, the **microstate of the system is represented by a point**. Furthermore, the temporal evolution of the system results in a trajectory within the phase space, traversing different microstates.

Example. Representation of an oscillation in the phase space.

$$F = -kx \implies m\ddot{x} = -kx \implies \ddot{x} + \frac{k}{m}x = 0$$

Calling $\omega^2 = k/m$ we get the following $x(t)$ and $p(t) = m\dot{x}(t)$

$$\begin{aligned} x(t) &= C_1 \cos(\omega t) + C_2 \sin(\omega t) \\ p(t) &= -m\omega C_1 \sin(\omega t) + m\omega C_2 \cos(\omega t) \end{aligned}$$

Using initial conditions, $x(t=0) = x_0$ and $p(t=0) = p_0$, we obtain the values of C_1 and C_2 which are $C_1 = x_0$ and $C_2 = p_0/m\omega$. Replacing these values in our functions we get.

$$\begin{aligned}x(t) &= x_0 \cos(\omega t) + p_0/m\omega \sin(\omega t) \\p(t) &= -m\omega x_0 \sin(\omega t) + p_0 \cos(\omega t)\end{aligned}$$

Multiplying $x(t)$ by $m\omega$, then square $x(t)$ and $p(t)$ and sums them we have

$$m^2\omega^2 x^2(t) + p^2(t) = m^2\omega^2 x_0^2 + p_0^2 = cte \quad (1.2)$$

Let's analyse the units of the previous equation and let's repeat it after multiply both sides of the equation by $1/2m$.

$$[m^2\omega^2 x^2(t)] = [kg \cdot m/s]^2, \quad [p^2(t)] = [kg \cdot m/s]^2$$

$$\frac{1}{2}m\omega^2 x^2(t) + \frac{1}{2m}p^2(t) = \frac{1}{2}m\omega^2 x_0^2 + \frac{1}{2m}p_0^2 = cte \quad (1.3)$$

$$[(m/2)\omega^2 x^2(t)] = [kg \cdot m^2/s^2] = [N \cdot m] = [J], \quad [(1/2m)p^2(t)] = [N \cdot m]$$

We can notice that we obtain units of **energy**.

In the phase space, the oscillation motion has a graphic of an ellipse with semiaxis $\sqrt{2mC'}$ and $\sqrt{2C'/m\omega^2}$ with C' the right side of the eq (1.3)

1.1.2. Dynamic functions

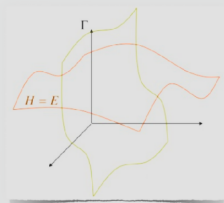
- There are some quantities that characterise the system globally. Some of these quantities can be measured or have a straightforward physical interpretation, such as energy, the system's momentum, or its angular momentum.
- These quantities have a defined value for each microstate $(\mathbf{q}, \mathbf{p})$; in other words, they depend on the dynamic state of the system. We will refer to this type of quantities as **dynamic functions**, and we will denote them as $a(\mathbf{q}, \mathbf{p})$.
- An important example is the **Hamiltonian** of the system: $H(\mathbf{q}, \mathbf{p})$. If it is an isolated and conservative system, then $H(\mathbf{q}, \mathbf{p})$ does not depend on time, and coincides with the total energy of the system, E .

In an isolated system, only microstates situated on the hypersurface characterised by the constant energy $H = E$ are accessible.

Accessible microstate
of the system

$\longleftrightarrow$

Point in phase space



In the field of statistical physics, numerous global properties are used to describe a system. Some of these properties are either directly measurable or possess straightforward physical interpretations. For instance, properties like energy, momentum, or angular momentum fall into this category. These properties assume specific values

for each microstate $(\mathbf{q}, \mathbf{p})$, indicating their dependence on the system's **dynamic state**. These types of properties are known as dynamic functions, typically denoted as $a(\mathbf{q}, \mathbf{p})$.

One significant example is the Hamiltonian of the system, often represented as $H(\mathbf{q}, \mathbf{p})$. In the case of an isolated and conservative system, $H(\mathbf{q}, \mathbf{p})$ remains independent of time and corresponds to the total energy of the system, denoted as E :

$$H(\mathbf{q}, \mathbf{p}) = E \quad (1.4)$$

In an **isolated** system, only microstates situated on the hypersurface characterised by the constant energy $H = E$ are accessible. In a classical mechanical system, which is both isolated and conservative, the Hamiltonian function $H(\mathbf{q}, \mathbf{p})$ represents a crucial dynamic quantity. In this context, $H(\mathbf{q}, \mathbf{p})$ signifies the total energy of the system and remains independent of time. It encapsulates the intricate interplay of positions $(\mathbf{q})$ and momenta $(\mathbf{p})$ of all particles within the system.

1.1.3. Temporal Evolution of a Microstate

$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q})$

Hamiltonian equations of motion

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

Initial state
 $q(0)$
 $p(0)$

$\xrightarrow{\quad}$

Final state
 $q(t)$
 $p(t)$

$\left. \begin{array}{l} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{array} \right\} \forall i = 1, \dots, s$

- Trajectories cannot leave the accessible phase space. Two or more trajectories never intersect each other: the laws of Mechanics are **deterministic**.
- If two systems are characterised by the same Hamiltonian H and are in the same initial microstate $(\mathbf{q}_0, \mathbf{p}_0)$, after a certain time t , both will be in the **same final microstate**:

$$(q_0, p_0) = (q'_0, p'_0) \Rightarrow (q_t, p_t) = (q'_t, p'_t)$$
- If at $t = 0$, one system is in a microstate $(\mathbf{q}_0, \mathbf{p}_0)$, and another system with the same Hamiltonian is in a different one $(\mathbf{q}'_0, \mathbf{p}'_0)$, then after a time t , the two will be in **different microstates**:

$$(q_0, p_0) \neq (q'_0, p'_0) \Rightarrow (q_t, p_t) \neq (q'_t, p'_t)$$
- Chaos: two systems with the same Hamiltonian prepared in two very similar microstates evolve microscopically in very different ways: **high sensitivity to initial conditions**.

Nota

Generally, the Hamiltonian can be decomposed as the contribution of kinetic energy and potential energy

$$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + V(\mathbf{q}) \quad (1.5)$$

To understand the concept of the temporal evolution of a microstate within a dynamic system, we need to explore how the state of a system changes over time as it transitions between different microstates. Consider a classical mechanical system composed of a multitude of particles, each with its unique position and momentum. The complete description of the system at any given moment is provided by specifying the positions q and momenta p of all these particles. This set of $2s$ variables defines a microstate of the system at that particular instant. Now, let's focus on how this microstate evolves over time. To do this, we need to introduce the concept of **equations of motion**. These equations, often derived from fundamental principles like Newton's laws of motion, govern how the positions and momenta of

particles change as a function of time. In classical mechanics, Hamilton's equations of motion play a central role. These equations are a set of first-order differential equations that describe how the positions q and momenta p evolve over time:

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i} \quad (1.6)$$

The solution to these equations of motion provides the trajectory of a microstate in the phase space Γ , which was introduced above. This trajectory represents how the microstate's positions and momenta change as time progresses. Understanding the temporal evolution of microstates is fundamental for predicting the macroscopic behaviour of a system. By considering the **ensemble** of all possible microstates and their trajectories, statistical physics enables us to make probabilistic predictions about macroscopic observables, such as temperature, pressure, and entropy. These predictions are essential for understanding the behaviour of gases, liquids, solids, and many other complex systems in the realm of physics.

1. Trajectories cannot leave the accessible phase space. Two or more trajectories never intersect each other: the laws of Mechanics are **deterministic**.
2. If two systems are characterised by the same Hamiltonian H and are in the same initial microstate (q_0, p_0) , after a certain time t , both will be in the **same final microstate**.

$$(q_0, p_0) = (q'_0, p'_0) \implies (q_t, p_t) = (q'_t, p'_t)$$

3. If at $t = 0$, one system is in a microstate (q_0, p_0) , and another system with the same Hamiltonian is in a different one (q'_0, p'_0) , then after a time t , the two will be in **different microstates**.

$$(q_0, p_0) \neq (q'_0, p'_0) \implies (q_t, p_t) \neq (q'_t, p'_t)$$

4. Chaos: two systems with the same Hamiltonian prepared in two very similar microstates evolve microscopically in very different ways: **high sensitivity to initial conditions**.

1.1.4. Temporal Evolution of a Dynamic Function

Let's consider a **dynamic function** $a(\mathbf{q}, \mathbf{p})$ that doesn't explicitly depend on time.
Despite its lack of explicit time dependence, this function can change over time as a result of the system's motion, given that $q_i = q_i(t)$ and $p_i = p_i(t)$

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial a}{\partial p_i} \frac{dp_i}{dt} \right) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \dot{q}_i + \frac{\partial a}{\partial p_i} \dot{p}_i \right)$$

and if we use the Hamiltonian equations:

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \equiv \{a, H\}$$

Poisson brackets

If the dynamic function depends explicitly on time $\rightarrow \frac{d}{dt} a(\mathbf{q}, \mathbf{p}; t) = \{a, H\} + \frac{\partial a}{\partial t}$

Let's consider a **dynamic function** $a(\mathbf{q}, \mathbf{p})$ that doesn't explicitly depend on time. Despite its lack of explicit time dependence, this function can change over time as a result of the system's motion, given that $q_i = q_i(t)$ and $p_i = p_i(t)$.

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial a}{\partial p_i} \frac{dp_i}{dt} \right) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \dot{q}_i + \frac{\partial a}{\partial p_i} \dot{p}_i \right) \quad (1.7)$$

and if we use the Hamiltonian equations:

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \equiv \{a, H\} \quad (1.8)$$

where the brackets in the right-hand-side of the above equation are referred to as **Poisson brackets**. If the dynamic function depends explicitly on time, then

$$\frac{d}{dt}a(\mathbf{q}, \mathbf{p}; t) = \{a, H\} + \frac{\partial a}{\partial t} \quad (1.9)$$

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \frac{da}{dt} = \{a, H\}$$

This equation is the most important equation in Classical Mechanics and includes the canonical equations of Hamilton as a particular case

$$\begin{aligned} \dot{q}_j = \{q_j, H\} &= \sum_{i=1}^s \left(\frac{\partial q_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial q_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = \frac{\partial H}{\partial p_j} \\ \dot{p}_j = \{p_j, H\} &= \sum_{i=1}^s \left(\frac{\partial p_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial p_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = -\frac{\partial H}{\partial q_j} \\ \dot{H} = \{H, H\} &= 0 \end{aligned}$$

H is basically a constant of motion. In general, if a certain dynamic function a satisfies the condition $\{a, H\} = 0$, then $a(q, p)$ is a **constant of motion** of the system.

The above equation $\dot{a}(\mathbf{q}, \mathbf{p}) = \{a, H\}$ can be considered the most important equation in Classical Mechanics and includes the canonical equations of Hamilton as a particular case.

$$\dot{q}_j = \{q_j, H\} = \sum_{i=1}^s \left(\frac{\partial q_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial q_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = \frac{\partial H}{\partial p_j} \quad (1.10)$$

$$\dot{p}_j = \{p_j, H\} = \sum_{i=1}^s \left(\frac{\partial p_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial p_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = -\frac{\partial H}{\partial q_j} \quad (1.11)$$

$$\dot{H} = \{H, H\} = 0 \quad (1.12)$$

H is basically a constant of motion. In general, if a certain dynamic function a satisfies the condition $\{a, H\} = 0$, then $a(\mathbf{q}, \mathbf{p})$ is a **constant of motion** of the system.

Some points of discussions:

1. It is important to note that classical mechanics provides a well-defined and detailed microscopic description based on Newton's laws. However, in general, we can only obtain a formal solution for the microscopic behaviour of the system due to:
 - a) The **impossibility of experimentally determining** the initial conditions of $\sim 10^{23}$ particles
 - b) The **mathematical and numerical impossibility of solving** $\sim 6 \times 10^{23}$ differential equations.
2. An interesting question arises: Even if we could solve the equations, how is it possible for thermodynamics, for example, to emerge from such dynamic complexity? Would such a solution provide true understanding? Sometimes, knowing the analytical solution to a complex problem does not necessarily mean understanding what is happening.
3. Microscopic Time Reversibility. The microscopic dynamic equations (Hamilton's equations) are temporally reversible. This is demonstrated by reversing the sign of all momenta and the sign of time in the Hamiltonian equations, and observing that they remain invariant. How could this process generate irreversible macroscopic laws?

1.2. Quantum Description



An important difference between Classical Mechanics and Quantum Mechanics is established by the **Uncertainty Principle**.

"The position and momentum of a particle cannot be simultaneously determined in a measurement with arbitrary precision."

$$\Delta q_i \cdot \Delta p_i \geq \frac{\hbar}{2} \quad \text{for all } i = 1, \dots, s$$

-- CONSEQUENCE --

A quantum microstate cannot be projected as a point in phase space; we must represent it as **cells with dimensions $\Delta q_i \Delta p_i$** that satisfy the Heisenberg Principle. All points within a cell are **quantum mechanically indistinguishable** and therefore correspond to the same quantum state.

Note
The Uncertainty Principle is **not** a limitation on our ability to measure accurately but reflects a **fundamental property** of quantum systems

Planck constant

$$\hbar = \frac{h}{2\pi} \simeq 1.0546 \times 10^{-34} \text{Js}$$

$$h = 6.62 \times 10^{-34} \text{Js}$$

One fundamental distinction between Classical Mechanics and Quantum Mechanics (QM) lies in the Uncertainty Principle, a concept introduced by Werner Heisenberg (1901 - 1976). This principle asserts that the precision with which we can simultaneously measure the position and momentum of a particle is **fundamentally limited**. In mathematical terms, it can be expressed as:

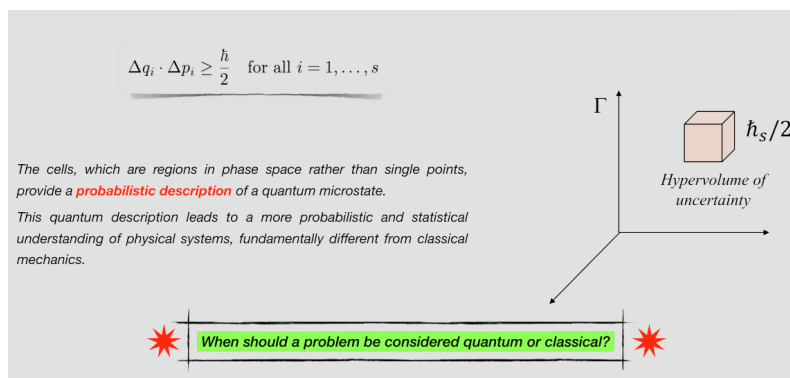
$$\Delta q_i \cdot \Delta p_i \geq \frac{\hbar}{2} \quad \text{for all } i = 1, \dots, s \quad (1.13)$$

where Δq_i represents the uncertainty in position, Δp_i signifies the uncertainty in momentum, and $\hbar = h/2\pi = 1.054571817 \dots 10^{-34} \text{ J s}$ (or, equivalently, J Hz^{-1})

denotes the reduced Planck constant, being h the Planck constant. This principle has profound implications for our understanding of the behaviour of particles at the quantum level. It implies that we cannot precisely know both the position and momentum of a particle simultaneously, no matter how advanced our measurement tools become. In a semiclassical view, this would correspond to having "fuzzy points" in phase space, with a volume proportional to $\hbar$.

Let's discuss some key features of this principle:

1. The Uncertainty Principle is a cornerstone of quantum theory and has led to a host of fascinating and counterintuitive phenomena, such as **wave-particle duality** and the **quantisation** of energy levels. It fundamentally challenges our classical intuition and underlines the inherent probabilistic nature of quantum systems.
2. The principle implies that there is an **inherent limit** to how precisely both the position and momentum of a particle can be known simultaneously. In other words, the more accurately we know the position of a particle, the less accurately we can know its momentum, and vice versa.
3. It fundamentally challenges classical physics, which assumes that it is possible to precisely determine both position and momentum. In QM, measurements are **inherently probabilistic**. Knowing the position of a particle precisely means that its momentum becomes highly uncertain, and vice versa.
4. The Uncertainty Principle is intimately connected to the concept of **wave-particle duality**. It implies that particles, such as electrons and photons, exhibit both particle-like and wave-like behaviour. The more precisely you try to determine their position, the more you disturb their momentum, and vice.
5. The Uncertainty Principle has wide-ranging implications in QM and plays a crucial role in the behaviour of subatomic particles. It has practical consequences in various quantum technologies, including quantum computing and quantum cryptography. versa.
6. Beyond its mathematical and physical significance, the Uncertainty Principle has profound philosophical implications. It challenges our classical intuitions about the determinism of physical systems and raises questions about the nature of reality at the quantum level.



In summary, the Uncertainty Principle is a foundational concept in QM, highlighting the fundamental limits on our ability to precisely know certain properties of particles. It underscores the probabilistic nature of quantum systems and has revolutionised our understanding of the behaviour of matter and energy at the smallest scales. An important consequence of the Uncertainty Principle is that a quantum microstate cannot be projected as a point in phase space. The classical description uses the concept of phase space, where each point represents a unique microstate of the system, specified by the generalised coordinates (q) and momenta (p). Instead, we must depict a quantum microstate as cells with dimensions $\Delta q_i \Delta p_i$ that satisfy Heisenberg's Uncertainty Principle. Consequently, within a cell in phase space, all points are quantum mechanically indistinguishable and correspond to the same quantum state. These **cells**, which are regions in phase space rather than single points, provide a probabilistic description of a quantum microstate. This quantum description leads to a more probabilistic and **statistical understanding of physical systems**, fundamentally different from classical mechanics. It is essential for understanding the behaviour of particles at the atomic and subatomic levels, where classical mechanics fails to provide an accurate description of their dynamics.

1.2.1. When should a problem be considered quantum or classical?

A problem must be addressed by **Quantum Mechanics** when the **action of the system is on the order of the Planck constant**

$$\mathcal{S} = \int_{t_1}^{t_2} \mathcal{L} dt \simeq h$$

Classical treatment will only be correct in the limit where the **action is much greater than the Planck constant**, allowing it to be considered that h is negligible

$$\mathcal{S} = \int_{t_1}^{t_2} \mathcal{L} dt \gg h$$

$\mathcal{L} = T - V$
Lagrangian

The smallest possible action is $h/2$; larger action values are integer multiples of this quantum

The distinction between a problem that should be treated quantum mechanically or classically depends on the scale and characteristics of the system studied. QM is fundamental when dealing with microscopic systems, such as atoms, molecules, and subatomic particles. At these scales, quantum effects dominate, and classical mechanics becomes inadequate to describe their behaviour accurately. Classical mechanics, on the other hand, works well for macroscopic systems, where quantum effects are negligible and the classical approximation is valid. More quantitatively, a problem must inevitably be **addressed using QM** when the **action of the system is on the order of Planck's constant**, $h = 6,62 \times 10^{-34}$ J s.

Ehrenfest's Theorem

Ehrenfest's Theorem states that a wave packet can be treated using Hamilton's equations (i.e., as a classical particle) if:

- $\Delta x \ll d$, where d is the typical distance traveled;
 - $\Delta p_x \ll \bar{p}$, average momentum of the particle (wave packet);
 - $\hbar/2 \leq \Delta x \Delta p_x \ll d \cdot \bar{p} \Rightarrow d \gg \hbar/2p$
-
- At high temperatures, thermal agitation causes the particles of a system (molecules, atoms, etc.) to move *over long distances* ($d \gg 0$) and possess *very large linear momenta* ($p \gg 0$). In other words, $d \gg \hbar/2p$, and we can apply the classical treatment.
 - At low temperatures, just the opposite happens: $d < \hbar/2p$, and it's necessary to apply QM.

Ehrenfest's Theorem

Ehrenfest's Theorem states that a wave packet can be treated using Hamilton's equations (i.e., as a classical particle) if:

1. $\Delta x \ll d$, where d is the typical distance traveled.
2. $\Delta p_x \ll \bar{p}$, average momentum of the particle (wave packet)
3. $\hbar/2 \leq \Delta x \Delta p_x \ll \bar{p} \Rightarrow d \gg \hbar/2p$

At high temperatures, thermal agitation causes the particles of a system (molecules, atoms, etc.) to move over long distances ($d \uparrow\uparrow$) and possess very large linear momenta ($p \uparrow\uparrow$). In other words, $d \gg \hbar/2p$, and we can apply the classical treatment. At low temperatures, just the opposite happens: $d < \hbar/2p$, and it's necessary to apply QM. **When can we assume that temperature is high enough to apply classical formalism?** There is no single answer to this question; it **depends on the type of system**. For a gas of atoms/molecules, room temperature (300 K) is already high enough to consider the system as classical. For a gas of electrons in a metal at room temperature, the system still exhibits strong quantum behaviour. For most solids at room temperature, classical behaviour holds true; at low temperatures (below 100 K), quantum effects become relevant. For liquid Helium II (which is achieved at temperatures very close to absolute zero), a quantum treatment is necessary.

1.2.2. State Representation in QM

- In the formalism of QM, the state of a system is represented by the **state vector** $|\Psi\rangle$, which encodes the **complete quantum state** of a physical system, including details about the positions, momenta, and other properties of its particles or components.
- $|\Psi\rangle$ serves as a mathematical representation that **enables predictions about the probabilities** of various outcomes when measurements are performed on the system.
- The **wave function** is the representation of $|\Psi\rangle$ in a given basis, **typically the position or momentum basis**, and its components are given by the projections of the state vector onto that basis.

$$\Psi(x, t) = \langle x | \Psi(t) \rangle$$

$$\Phi(p, t) = \langle p | \Psi(t) \rangle$$

$$|\Psi(t)\rangle = \int \Psi(x, t) |x\rangle dx$$

$$|\Psi(t)\rangle = \int \Phi(p, t) |p\rangle dp$$

This equation essentially says that the state vector $|\Psi\rangle$ is a superposition of all possible position eigenstates weighted by the corresponding values of the wave function $\Psi(x, t)$.

Relations state vector - wave function in bra-ket (Dirac) notation

In the formalism of QM, the state of a system is represented by a vector $|\Psi\rangle$ that encodes the **complete quantum state** of a physical system, including details

about the positions, momenta, and other properties of its particles or components. It serves as a mathematical representation that **enables predictions about the probabilities** of various outcomes when measurements are performed on the system. At a particular instant of time, the components of this vector represent the values of the corresponding **wave function**

$$\Psi = \Psi(q_1, \dots, q_s, t) \quad (1.14)$$

where s is the number of degrees of freedom. The link between state vector and wave function, due to the extremely large number of represented values, is better expressed as an integral. In bra-ket (Dirac) notation, this vector is written as

$$|\Psi(t)\rangle = \int \Psi(x, t) |x\rangle dx \quad (1.15)$$

By manipulating and evolving the wave function using quantum operators, one can make predictions about the behaviour of quantum systems, including particles like electrons, atoms, and molecules. The set of all states or wave functions possesses the structure of a **Hilbert space** $\hat{H}$, which is a complex vector space with an inner product that allows for the mathematical treatment of quantum states, operations, and measurements. This Hilbert space structure provides the mathematical framework for the description of quantum states and their evolution over time.

The main difference between phase space in classical mechanics and configuration space in QM lies in how they represent the state of a physical system and how variables are handled. Let's look at this difference in more detail.

Phase Space in Classical Mechanics			Configuration Space in QM	
- State Representation: Points in classical <i>phase space</i> represent the state of a physical system, characterised by position and momentum coordinates. - Classical Variables: Variables are <i>real numbers</i>, i.e., position and momentum coordinates are continuous and can take any real value. - Determinism: Classical phase space follows <i>deterministic trajectories</i> based on initial conditions, governed by Hamilton's or Newton's equations.			- State Representation: QM employs <i>configuration space</i> to represent the system's state, described by a wave function or state vector indicating the probability of finding a particle in a specific configuration. - Quantum Variables: Variables are <i>operators</i> that act on wave functions. - Probabilistic Nature: Quantum configuration space involves wave functions assigning <i>probabilities</i> to finding particles in specific configurations, with quantum predictions relying on probabilities and superpositions of states.	
Framework	Space	Dimension	Each axis of configuration space corresponds to one independent generalized coordinate, i.e., one degree of freedom of the total system. If the state vector is represented on the momentum basis, then the wave function is represented on momentum space, not configuration space.	
Classical configuration space	q_i	s		
Classical phase space	(q_i, p_i)	$2s$		

Phase Space in Classical Mechanics

1. **State Representation:** In classical mechanics, phase space represents the state of a physical system. Each point in phase space corresponds to a unique configuration of the system, defined by the coordinates of position and momentum of all particles in the system.

2. **Classical Variables:** In classical mechanics, the variables that describe the state are real numbers, i.e., position and momentum coordinates are continuous and can take any real value.
3. **Determinism:** In classical phase space, the system follows deterministic trajectories. Given an initial set of initial conditions, the system evolves uniquely along a specific trajectory in phase space, according to Hamilton's equations or Newton's equations of motion.

Configuration Space in QM

1. **State Representation:** In QM, configuration space represents the state of the system. A quantum state is described by a wave function or a state vector that contains information about the probability of finding a particle in a particular configuration.
2. **Quantum Variables:** In QM, the variables that describe the state are operators that act on wave functions. These operators generally do not commute, leading to properties such as non-commutativity and Heisenberg's uncertainty.
3. **Probabilistic Nature:** In quantum configuration space, the behaviour of particles is described through wave functions that assign probabilities of finding a particle in a specific configuration. Quantum predictions are related to probabilities and superpositions of states.

Nota

Each axis of configuration space corresponds to **one independent generalized coordinate**, i.e., one degree of freedom of the total system. If the state vector is represented on the momentum basis, then the wave function is represented on **momentum space**, not configuration space.

In summary, while phase space in classical mechanics is based on continuous variables and allows for deterministic evolution of the system, configuration space in QM is based on wave functions and non-commutative operators, and predictions are related to **probabilities** and the **principle of superposition**. These fundamental differences reflect the essential distinction between classical physics and quantum physics.

In QM, one of the fundamental concepts is the probability of finding a physical system in a particular state within its configuration space. This probability is described by the wave function $\Psi(\mathbf{q}, t)$, where $\mathbf{q}$ represents the configuration coordinates of the system and t time. The probability $P(\mathbf{q}, t)$ of finding the system within the region of the configuration space defined by $[q_1, q_1 + dq_1][q_2, q_2 + dq_2] \dots$ at a specific time t reads

$$P(\mathbf{q}, t) = \int_q^{q+dq} |\Psi(\mathbf{q}, t)|^2 d\mathbf{q}$$

where $\Psi(\mathbf{q}, t)$ is the wave function of the system in the configuration space $\mathbf{q}$ at time t ; $|\Psi(\mathbf{q}, t)|^2$ is the square modulus of the wave function, providing the probability density; and $d\mathbf{q} = dq_1 \dots dq_s$ represents the infinitesimal volume element in the configuration space.

In QM, one of the fundamental concepts is the probability of finding a physical system in a particular state within its configuration space. This probability is described by the wave function $\Psi(\mathbf{q}, t)$, where $\mathbf{q}$ represents the configuration coordinates of the system and t time. The probability $P(\mathbf{q}, t)$ of finding the system within the region of the configuration space defined by $[q_1, q_1 + dq_1][q_2, q_2 + dq_2] \dots$ at a specific time t reads:

$$P(\mathbf{q}, t) = \int_q^{q+dq} |\Psi(\mathbf{q}, t)|^2 d\mathbf{q} \quad (2.18)$$

where $\Psi(\mathbf{q}, t)$ is the wave function of the system in the configuration space $\mathbf{q}$ at time t ; $|\Psi(\mathbf{q}, t)|^2$ is the square modulus of the wave function, providing the probability density; and $d\mathbf{q} = dq_1 \dots dq_s$ represents the infinitesimal volume element in the configuration space.

Example 2.2: Probability of Finding a Particle in a Specific Range

Consider a one-dimensional quantum system with the wave function $\Psi(x, t)$. We want to calculate the probability of finding the particle between $x = 2$ and $x = 4$ at $t = 0$. According to QM, the probability $P(2 \leq x \leq 4; 0)$ of finding the particle in the range $[2, 4]$ at time $t = 0$ is given by:

$$P(2 \leq x \leq 4; 0) = \int_2^4 |\Psi(x, 0)|^2 dx \quad (1.16)$$

Suppose the wave function is given by $\Psi(x, 0) = 2\sqrt{2}e^{-x^2}$. Plugging this into the formula, we can calculate the probability:

$$P(2 \leq x \leq 4; 0) = \int_2^4 |2\sqrt{2}e^{-x^2}|^2 dx \quad (1.17)$$

Now, compute the integral:

$$P(2 \leq x \leq 4; 0) = \int_2^4 8e^{-2x^2} dx \quad (1.18)$$

This integral can be evaluated numerically to find the probability of locating the particle between $x = 2$ and $x = 4$ at $t = 0$. The result will be a numerical value $P(2 \leq x \leq 4; 0) \approx 6,27 \cdot 10^{-5}$. But, is this a probability? If the probability in the full space is equal to one, then the answer is yes, else we have to normalise. The integral of the full space give us $\sim 10,026$, then the probability is $6,27 \cdot 10^{-5} / 10,026 \approx 6,25 \cdot 10^{-6}$.

1.2.3. Wave function and the Schrödinger equation

The state vector depends on time, and its evolution is given by the **Schrödinger equation**

General form of the Schrödinger equation including time dependence

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = \hat{H} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$$

$\hat{H}$ represents the Hamiltonian operator describing the total energy of the system.

- ✓ The state vector depends on time.
- ✓ Its evolution is governed by the Schrödinger equation.
- ✓ The evolution is unitary if the Hamiltonian is self-adjoint.
- ✓ In position space, this becomes the time-dependent wave equation for $\psi(\mathbf{r}, t)$.

Unitary evolution: as the quantum system evolves in time, the total probability of all possible outcomes stays exactly 1.

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For a system with constant volume V and a fixed number of particles N , the **Schrödinger equation** must be satisfied:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = \hat{H} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) \quad (1.19)$$

where $\hat{H}$ represents the Hamiltonian operator describing the total energy of the system; $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$ is the wave function of the system, dependent on the coordinates $\mathbf{r}_i$ of the particles and time t ; i is the imaginary unit; and $\hbar$ is the reduced Planck constant. This equation is the fundamental form of the Schrödinger equation in QM, governing the evolution of the wave function for a quantum system with N particles in a volume V .

$\hat{H}\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$

This equation is employed only when the **Hamiltonian does not explicitly depend on time**.

Nevertheless, even in this scenario, the overall wave function remains time-dependent.

The solutions of this equation are the vectors $|\psi_{E_n}\rangle$, in the basis of energy eigenstates, whose linear combination gives the **time-dependent state vector** $|\Psi(t)\rangle$:

$$|\Psi(t)\rangle = \sum_n A_n e^{-iE_n t/\hbar} |\psi_{E_n}\rangle$$

★

A_n : complex numbers

time-evolution operator (propagator)

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Additionally, if the energy E of the system is **constant** (an isolated system), then the above equation simplifies, and the time dependence disappears:

$$\hat{H}\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \quad (1.20)$$

The above equation is employed only when the Hamiltonian itself does not explicitly depend on time. Nevertheless, even in this scenario, the overall wave function remains time-dependent. In particular, the solutions of this equation are the vectors

ψ_{E_n} , in the basis of energy eigenstates, whose linear combination gives the following **time-dependent state vector**:

$$|\Psi(t)\rangle = \sum_n A_n e^{-iE_n t/\hbar} |\psi_{E_n}\rangle \quad (1.21)$$

where A_n are complex numbers and $e^{-iE_n t/\hbar}$ represents the time evolution operator, also known as the **propagator**, which describes the time evolution of a quantum state under a time-independent Hamiltonian.

Due to the principle of uncertainty, even if the wave function is known, it is not possible to uniquely determine all the values of q_i and p_i : the wave function Ψ determines the probability distribution of physical quantities but not their exact values. More specifically, the square modulus of the wave function $|\Psi|^2$ provides information about the probability density of finding a particle in a particular state or with specific values of physical quantities such as position or momentum. However, it does not provide the exact values of these quantities. This probabilistic nature of QM is a fundamental feature, and it means that we can predict the likelihood of different outcomes, but we cannot precisely determine individual values. The wave function encodes the probabilities associated with various outcomes, reflecting the inherent uncertainty and indeterminacy at the quantum level.

1.2.4. Pure States and Mixed States

◦ If we are certain that the state of the quantum system is in state $|\psi\rangle$, then it is said to be in a **pure state**.

◦ If the system can be in various quantum states, and we do not know which one it is, then it is considered a **mixed state**.

PURE STATE

When a system is in a pure state, all information about its quantum properties is contained in a **single** state vector. Pure states are described by wave functions that are not statistical mixtures of other states.

$$\hat{\rho}_{\text{pure}} = |\psi\rangle\langle\psi|$$

MIXED STATE

Mixed states result from a *statistical mixture* of pure states. In a mixed state, the system has *uncertainty or lack of knowledge* about its quantum properties. They are represented by **density operators** or **density matrices**, which encode all measurable information

$$\hat{\rho}_{\text{mixed}} = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad \sum_i p_i = 1$$

WHAT IS AN OPERATOR????

In QM, there are two fundamental types of quantum states.

- **Pure states** are represented by vectors $|\Psi\rangle$ in a Hilbert space $\hat{H}$. These states are characterised by having a well-defined and definite quantum state. When a system is in a pure state, all information about its quantum properties is contained in a **single** state vector. Pure states are described by wave functions that are not statistical mixtures of other states.
- **Mixed states**, on the other hand, are represented by **density operators** or **density matrices** ρ . These states result from a statistical mixture of pure states. In a mixed state, the system has uncertainty or lack of knowledge about its quantum properties. The density operator ρ contains information about the probabilities of finding the system in different pure states. Mixed states arise in

situations where there is incomplete information or when a system is prepared probabilistically.

Understanding and distinguishing between pure states and mixed states is essential in QM as they have different mathematical descriptions and physical interpretations. Pure states correspond to definite quantum states, while mixed states represent statistical ensembles of possible pure states.

1.2.5. Association of physical quantities with quantum operators

In QM, operators encode both the possible values of the observable and how the system responds to measurement.
Physical observables are represented by special operators called **self-adjoint operators**.

When applied to a ket $|\psi\rangle$, an operator **transforms it into a new ket** $|\psi'\rangle$ within the same space (potentially a scaled version of the original ket).

$$\hat{B}|\psi\rangle = |\psi'\rangle$$

the same operator can also transform bras

$$\langle\psi|\hat{B} = \langle\psi'|$$

Self-adjoint operator $\hat{B}$: it is equal to its own adjoint (conjugate transpose)

$$\hat{B} = \hat{B}^\dagger$$

Hamiltonian ($\hat{H}$): Applying the Hamiltonian operator (which looks different for every different physical situation) gives you E , the energy of the particle represented by the ket $|\psi\rangle$; E is a scalar quantity.

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

Unity or identity ($\hat{I}$): The unity or identity operator leaves kets unchanged:

$$\hat{I}|\psi\rangle = |\psi\rangle$$

Gradient (∇): The gradient operator works like this:

$$\nabla|\psi\rangle = \frac{\partial}{\partial x}|\psi\rangle + i\frac{\partial}{\partial y}|\psi\rangle + j\frac{\partial}{\partial z}|\psi\rangle + \dots$$

Linear momentum ($\hat{P}$): The linear momentum operator looks like this in quantum mechanics:

$$\hat{P}|\psi\rangle = -i\hbar\nabla|\psi\rangle$$

Laplacian (∇^2): You use the Laplacian operator, which is much like a second-order gradient, to create the energy-finding Hamiltonian operator:

$$\nabla^2|\psi\rangle = \nabla \cdot \nabla|\psi\rangle = \frac{\partial^2}{\partial x^2}|\psi\rangle + \frac{\partial^2}{\partial y^2}|\psi\rangle + \frac{\partial^2}{\partial z^2}|\psi\rangle + \dots$$

$\hat{B} = \hat{B}^\dagger \quad \rightarrow \quad \langle\psi_1|\hat{B}|\psi_2\rangle = \langle\psi_2|\hat{B}|\psi_1\rangle^* \quad \text{Symmetry of the inner product} \quad \star$

In the context of QM, each physical quantity B is associated with a self-adjoint linear operator $\hat{B}$ that operates on the states within the Hilbert space.

1. **Physical Quantity B :** In the context of QM, B represents a physical quantity or observable, such as position, momentum, energy, or any other property that we want to measure or describe.
2. **Self-adjoint Linear Operator $\hat{B}$:** This refers to a mathematical operator denoted as $\hat{B}$ that acts on the quantum states within the Hilbert space. The term “**self-adjoint**” means that the operator $\hat{B}$ is Hermitian, which implies that it is equal to its own adjoint (or conjugate transpose). Self-adjoint operators are crucial in QM because they correspond to observable physical quantities, and they have real eigenvalues.
3. **Hilbert Space:** The Hilbert space is a mathematical space used in QM to describe the possible states of a quantum system. It is a complex vector space where quantum states, represented as kets (e.g., $|\psi\rangle$), reside. These states can be operated upon by quantum operators like $\hat{B}$ to obtain measurements or predictions of physical quantities.

Properties of the self-adjoint operator

A self-adjoint operator $\hat{B}$ in QM possesses several key properties:

1. **Hermitian Nature:** A self-adjoint operator $\hat{B}$ is Hermitian, which means it is equal to its own adjoint or conjugate transpose:

$$\hat{B} = \hat{B}^\dagger \quad (1.22)$$

This property ensures that the inner product (expected value) of $\hat{B}$ acting on a state vector is the same whether one takes the inner product of ψ_1 and $\hat{B}\psi_2$ or ψ_2 and $\hat{B}\psi_1$ (symmetry of the inner product). Mathematically, $\langle \psi_1 | \hat{B} | \psi_2 \rangle = \langle \psi_2 | \hat{B} | \psi_1 \rangle^*$.

2. **Real Eigenvalues:** The eigenvalues of a self-adjoint operator $\hat{B}$ are always real numbers. This property is crucial because it ensures that measurements of physical quantities associated with $\hat{B}$ yield real values.
1. **Orthogonal Eigenvectors:** The eigenvectors corresponding to distinct eigenvalues of $\hat{B}$ are orthogonal to each other. In other words, if two eigenvalues λ_i and λ_j are different, then the eigenvectors $|i\rangle$ and $|j\rangle$ associated with these eigenvalues are orthogonal:

$$\langle i | j \rangle = 0 \quad \text{for } \lambda_i \neq \lambda_j \quad (1.23)$$

2. **Completeness:** The set of eigenvectors of $\hat{B}$ forms a complete orthonormal basis for the Hilbert space. This property allows any quantum state to be expressed as a linear combination of these eigenvectors.

1.2.6. Expectation value

Expectation value of B or quantum average

$$\bar{B} = \langle \psi | \hat{B} | \psi \rangle = \langle \psi | (\hat{B} \psi) \rangle = \int \psi^* (\hat{B} \psi) dq_1 \dots dq_n \quad |\psi\rangle \text{ normalised to 1, that is } \langle \psi | \psi \rangle = 1$$

The expectation value of an operator is the **average value** that one would measure if the measurement was performed many times (under identical conditions!).

Because $\langle \psi |$ can be expressed as a row operator and $|\psi\rangle$ as a column vector, $\hat{B}$ takes the form of as a **square matrix**.

In QM, the numerical value observed in a measurement of a certain physical quantity after preparing the system in a specific microstate ψ is not defined. The only quantity we can define is its average value. This average value is what we would obtain after performing a large number of measurements under identical conditions (the same microstate ψ). This is usually referred to as **quantum average**. The result of a measurement (observable) for a quantum system in the state ψ is equal to the expected value (average) of the operator $\hat{B}$ for that state ψ . In mathematical terms, this can be expressed as:

$$\bar{B} = \langle \psi | \hat{B} | \psi \rangle = \langle \psi | (\hat{B} \psi) \rangle = \int \psi^* (\hat{B} \psi) dq_1 \dots dq_n \quad (1.24)$$

This equation reflects the fundamental concept that in QM, the result of a measurement is probabilistic, and the expectation value of the operator $\hat{B}$ provides the average or expected outcome for that measurement when the system is in the state ψ . We also notice that $|\psi\rangle$ is normalised to unity, that is $\langle\psi|\psi\rangle = 1$. This condition guarantees that the probability of finding the system in any possible state is conserved and equal to 1. Finally, one should note that the fact that $\hat{B}$ is self-adjoint ensures that its expected value is a real number.

When our quantum system is in a mixed state, meaning it can exist in multiple quantum states with certain probabilities, the outcome of a measurement is probabilistic. In this scenario, *we cannot predict a single definite value for the measurement*. Instead, we can calculate the expected or average value of the measurement, which is known as the “expectation value.” The expectation value represents the average outcome we would obtain if we were to repeat the same measurement on identically prepared systems in their respective mixed states many times. It is a fundamental concept in QM and is calculated using the principles of quantum probability.

1.2.7. Eigenvalues, eigenstates, eigenfunctions

An **eigenfunction** of an observable (position, momentum, energy, etc.) is a **state** for which the measurement of that observable yields a definite value (the **eigenvalue**, a real number).

When we measure a physical observable, the only possible outcome of this measurement is one of the **eigenvalues** of the associated operator (always real because operators are self-adjoint)

After a measurement that yields a given eigenvalue, the system is left in a state belonging to the corresponding **eigenspace** of the operator (**projection postulate**).

Ψ is an eigenfunction
of operator $\hat{\Omega}$

Eigenvalue equation

$$\hat{\Omega}\Psi = \omega\Psi$$

ω is an eigenvalue
of operator $\hat{\Omega}$

(see also slide)

(OPERATOR) (FUNCTION) = (CONSTANT) (SAME FUNCTION)

The value of the observable for the system is the eigenvalue, and the system is said to be in an **eigenstate**

The eigenvalue equation in QM tells us that measurable physical quantities correspond to operators, and their eigenvalues represent possible definite measurement outcomes.

Physical Interpretation

- Eigenfunctions allow one to **extract measurement information** from operators.
- Eigenvalues are **the real numbers that correspond to what one observes experimentally**.
- Measurement changes the system: after the measurement, it is in the **eigenspace corresponding to the measured value**, guaranteeing repeatability if we measure immediately again.

In QM, eigenfunctions are special functions associated with a physical observable (represented by the operator $\hat{B}$) that have the property that when the operator $\hat{B}$ acts on them, it returns a scaled version of the same function. Mathematically, this can be expressed as:

$$\hat{B}|\phi_n\rangle = b_n|\phi_n\rangle \quad (1.25)$$

where $\hat{B}$ is the operator corresponding to the physical observable and ϕ_n represents the *eigenfunction* associated with the *eigenvalue* b_n . This equation is usually referred to as the **eigenvalue equation**. It describes how certain special states

(called *eigenstates*) of an operator remain the same except for a scaling factor (the eigenvalue) when acted upon by that operator.

Eigenfunctions are crucial in QM because they provide a basis in which any quantum state can be expressed through a linear combination. This means that if we know the eigenfunctions of an operator, we can represent any quantum state using those eigenfunctions and coefficients. Decomposition of a state vector into an orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is a fundamental concept in vector space theory and also applies in QM.

Example:

Eigenvalue equation

(OPERATOR) (FUNCTION) = (CONSTANT) (SAME FUNCTION) $\Rightarrow \hat{\Omega}\Psi = \omega\Psi$

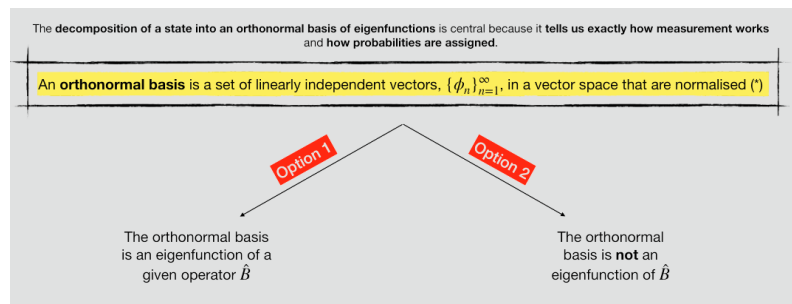
Example: are e^{ax} and e^{ax^2} **eigenfunctions** of the operator $\hat{\Omega} \equiv \frac{d}{dx}$???

$\Psi = e^{ax}$ and $\Omega = \frac{d}{dx} \Rightarrow \hat{\Omega}\Psi = \omega\Psi \Leftrightarrow \frac{d}{dx}(e^{ax}) = a e^{ax} = \omega\Psi$ ✓

$\Psi = e^{ax^2}$ and $\Omega = \frac{d}{dx} \Rightarrow \hat{\Omega}\Psi = \omega\Psi \Leftrightarrow \frac{d}{dx}(e^{ax^2}) = 2ax e^{ax^2} \neq \omega\Psi$ ✗

↑
Not a constant!

1.2.8. Decomposition of $|\psi\rangle$ into an orthonormal basis



In this context, an orthonormal basis is a set of linearly independent vectors in a vector space that are normalised, meaning they have a norm (length) equal to 1 and are mutually orthogonal (the inner product between two different vectors is 0). In QM, the states of a system are represented as vectors in a Hilbert space, and these states can be decomposed in terms of an orthonormal basis. There are two options here: either the orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is an eigenfunction of $\hat{B}$ or it is not.

Case I. The orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is an eigenfunction of $\hat{B}$.

Option 1 $\hat{B}|\phi_n\rangle = b_n|\phi_n\rangle$ Each value b_n corresponds to a possible value of the observable B .
elements of the orthonormal basis (eigenfunctions of operator $\hat{B}$)

The pure quantum state ψ can be decomposed in this orthonormal basis $\Rightarrow |\psi\rangle = \sum_{n=1}^{\infty} c_n|\phi_n\rangle$

$c_n = \langle\phi_n|\psi\rangle = \int \phi_n^* \psi dq_1 \dots dq_N$

$|c_n|^2$ is the probability that, given a system in the pure state $|\psi\rangle$, after a measurement it will be found in the state $|\phi_n\rangle$ $\longrightarrow \sum_{n=1}^{\infty} |c_n|^2 = 1$

The expected value of a certain magnitude is obtained as an average over all possible values b_n . $\Rightarrow \bar{B} = \langle\psi|\hat{B}|\psi\rangle = \sum_n |c_n|^2 b_n$ ★

The decomposition of a quantum state $|\psi\rangle$ in terms of the eigenfunctions of the operator $\hat{B}$ can be written as:

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle \quad (1.26)$$

where $|\psi\rangle$ is the quantum state we want to decompose; $|\phi_n\rangle$ are the elements of the orthonormal basis (eigenfunctions of operator $\hat{B}$); and c_n are complex numbers describing how much each element of the basis contributes to the state $|\psi\rangle$. To find the coefficients c_n , inner products and properties of the orthonormal basis are used. In particular, c_n is calculated as the inner product between the state $|\psi\rangle$ and the basis element $|\phi_n\rangle$:

$$c_n = \langle\phi_n|\psi\rangle = \int \phi_n^* \psi dq_1 \dots dq_N \quad (1.27)$$

Additionally

$$\langle\psi| = \sum_{n=1}^{\infty} \langle\phi_n| c_n^* \quad (1.28)$$

where $|c_n|^2$ is the probability that, given a system in the pure state $|\psi\rangle$, after a measurement it will be found in the state $|\phi_n\rangle$. It must hold that

$$\sum_{n=1}^{\infty} |c_n|^2 = 1 \quad (1.29)$$

It follows that the expected value of the observable $\hat{B}$ reads

$$\bar{B} = \langle\psi|\hat{B}|\psi\rangle = \left(\sum_{m=1}^{\infty} \langle\phi_m| c_m^* \right) \hat{B} \left(\sum_{n=1}^{\infty} c_n |\phi_n\rangle \right) = \sum_{m=1}^{\infty} c_m^* \sum_{n=1}^{\infty} c_n \underbrace{\langle\phi_m|\hat{B}|\phi_n\rangle}_{b_n \delta_{nm}} \quad (1.30)$$

where $|\phi_n\rangle$ are eigenvalues of $\hat{B}$. This equation reduces to

$$\bar{B} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n b_n \langle\phi_m|\phi_n\rangle = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n b_n \delta_{nm} = \sum_{n=1}^{\infty} |c_n|^2 b_n \quad (1.31)$$

where $\delta_{nm} \equiv \langle \phi_m | \phi_n \rangle$ is an orthonormal basis and b_n is the value of B in state n (real number). In other words, $\bar{B}$ is an average, where each b_n should be weighted by its corresponding probability $|c_n|^2$.

Case II. The orthonormal basis $\{\phi_n\}_{n=1}^{\infty}$ is not an eigenfunction of $\hat{B}$.

Option 2

$$\hat{B}|\phi_n\rangle \neq b_n|\phi_n\rangle$$

$$\bar{B} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n \underbrace{\langle \phi_m | \hat{B} | \phi_n \rangle}_{b_{mn}}$$

Expectation value of B

b_{mn} is an element of matrix (m, n) and can be a complex number.

$\bar{B}$ is a real number, but its expression is not that typical of an average:
it is a **quantum average** and its contributions b_{mn} are consequence of **quantum interferences**: they do not have a classical analogue.

In this case,

$$\hat{B}|\phi_n\rangle \neq b_n|\phi_n\rangle \quad (1.32)$$

and

$$\bar{B} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} c_m^* c_n \underbrace{\langle \phi_m | \hat{B} | \phi_n \rangle}_{b_{mn}} \quad (1.33)$$

where b_{mn} is an element of matrix (m, n) and can be a complex number. We conclude that $\bar{B}$ is a real number, but its expression is not that typical of an average: it is a quantum average and its contributions b_{mn} are consequence of quantum interferences: they do not have a classical analogue.

Generally speaking, decomposition into an orthonormal basis is fundamental for understanding and working with quantum systems, as it allows us to express any state in terms of well-defined basic states and calculate probabilities and expectations of observables. Additionally, it ensures that the norm of the state $|\psi\rangle$ is preserved in the decomposition.

1.2.9. Temporal Evolution of an Operator

$$i\hbar \frac{\partial}{\partial t} |\psi_t\rangle = \hat{H} |\psi_t\rangle \quad \longrightarrow \quad \frac{d\hat{B}_t}{dt} = \frac{1}{i\hbar} [\hat{B}_t, \hat{H}] \quad \text{Commutator}$$

Remember that in Classical Mechanics, the temporal evolution of a dynamic function reads

$$\dot{a}(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^s \left(\frac{\partial a}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial a}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \equiv \{a, H\}$$

$$\hat{B}_t \hat{H} - \hat{H} \hat{B}_t$$

Show that momentum and position do NOT commute ★

Solution on PRADO

Classical Mechanics Quantum Mechanics

$\{\cdot, \cdot\}$ $\longleftrightarrow$ $\frac{1}{i\hbar} [\cdot, \cdot]$

.....

An observable quantity will be a **constant of motion** if it commutes with the Hamiltonian and it does not itself depend explicitly on time

The temporal evolution of states $|\psi_t\rangle \in \hat{H}$ follows the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi_t\rangle = \hat{H} |\psi_t\rangle \quad (1.34)$$

This equation has a formal solution. In the case where the Hamiltonian is independent of time:

$$|\psi_t\rangle = e^{-\frac{i\hat{H}t}{\hbar}} |\psi(0)\rangle \quad (1.35)$$

where $|\psi(0)\rangle$ is the wave function at $t = 0$ (independent from time). Additionally

$$\langle \psi_t | = \langle \psi(0) | e^{\frac{i\hat{H}t}{\hbar}} \quad (1.36)$$

The expected value of an operator $\hat{B}$ at time t reads

$$\langle \psi_t | \hat{B} | \psi_t \rangle = \langle \psi(0) | \underbrace{e^{\frac{i\hat{H}t}{\hbar}} \hat{B} e^{-\frac{i\hat{H}t}{\hbar}}}_{\hat{B}_t} | \psi(0) \rangle = \langle \psi(0) | \hat{B}_t | \psi(0) \rangle \quad (1.37)$$

where we have transferred the time dependence to the operator $\hat{B}_t$. It follows that we can calculate the derivative of $\hat{B}_t$ with respect to time:

$$\frac{d\hat{B}_t}{dt} = \frac{d}{dt} \left(e^{\frac{i\hat{H}t}{\hbar}} \hat{B} e^{-\frac{i\hat{H}t}{\hbar}} \right) = \frac{i}{\hbar} \hat{H} e^{\frac{i\hat{H}t}{\hbar}} \hat{B} e^{-\frac{i\hat{H}t}{\hbar}} - \frac{i}{\hbar} e^{\frac{i\hat{H}t}{\hbar}} \hat{B} \hat{H} e^{-\frac{i\hat{H}t}{\hbar}} = \frac{i}{\hbar} [\hat{H} \hat{B}_t - \hat{B}_t \hat{H}] = \frac{1}{i\hbar} [\hat{B}_t \hat{H} - \hat{H} \hat{B}_t] \quad (1.38)$$

Equivalently:

$$\frac{d\hat{B}_t}{dt} = \frac{1}{i\hbar} [\hat{B}_t, \hat{H}] \quad (1.39)$$

If $[\hat{B}_t, \hat{H}] = 0$, then $\hat{B}_t$ is a constant of motion of the system⁵ and there exists a basis that diagonalises both operators simultaneously.